

Exploratory Data Analysis Report

1. Dataset Overview

Total Rows: 720

Total Columns: 21

Columns Included:

year, month, day, hour, PM10, PM2.5, AT (Air Temp), BP (Barometric Pressure), SR (Solar Radiation), RH (Humidity), WD (Wind Direction), NO, NO2, SO2, Ozone, CO, Benzene, NH3, NOx, and an index column.

2. Time-based Trends

The data spans a single month (June 2018), over 30 days, with hourly frequency. A timestamp was created by combining year, month, day, and hour for trend analysis.

PM2.5 over Time:

PM2.5 values showed natural diurnal variation.

Peaks are seen in early mornings and late evenings.

Relatively lower values during mid-day hours.

3. Correlation Matrix

Strong positive correlation between:

NO2 and NOx

PM10 and PM2.5

CO and Benzene

Negative correlation observed between Ozone and NO.

4. Outlier Detection (Visual)

PM2.5 has a few values beyond 120, which could be flagged as potential pollution spikes.

Benzene and NH3 also show scattered high values suggesting occasional industrial impact.

5. Key Observations

PM10 levels are consistently high, suggesting poor air quality.

Ozone levels vary significantly, but are negatively correlated with NO and NO2, aligning with known atmospheric chemistry.

CO, Benzene, and NH3 values hint at urban or industrial air pollution.

6. Summary Statistics

PM2.5

Mean: ~76.65

Range: 35 – 146

PM10

Mean: ~255.48

Range: 115 – 365

Temperature (AT): ~33.02°C

Humidity (RH): ~69.8%

